Code No: R07A1EC10 **R07** **Set No. 2**

I B.Tech Examinations,May 2011
COMPUTER PROGRAMMING AND NUMERICAL METHODS
Metallurgy And Material Technology

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

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1. (a) Write an example to explain about call by value.
   (b) Write a factorial program using no return and no argument type. [10+6]
2. Define ADT. Explain with an example. [16]
3. (a) Using Lagrange's formula find f(6) given

X:	2	5	7	10	12
F(x)	18	180	448	1210	2028

   (b) If the interval of differencing is unity, prove that $\Delta \sin x = 2\sin {}^1\!/_2 \cos\left(x + {}^1\!/_2\right)$. [8+8]
4. (a) Find the positive root x^3- x = 1 correct to four decimal places by bisection method.
   (b) Find the parabola of the form y = ax^2 + bx + c passing through the points (0, 0), (1, 1) and (2, 2). [8+8]
5. What is the purpose of iterative statements? Explain about for loop with an example. [16]
6. (a) Write short notes on pointer to void.
   (b) Write short notes on Address Arithmetic. [8+8]
7. (a) By dividing the range in to five equal parts, evaluate $\int_0^\pi \sin x dx$ by Trapezoidal rule and Simpson's rule.
   (b) Using Milne's method find y (4.4) given $5xy' + y^2 - 2 = 0$ given y (4) =1, y(4.1) =1.0097, y(4.3) = 1.0143. [8+8]
8. (a) How can a entire structure be passed to a function?
   (b) How can a entire structure be returned from a function? [8+8]

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Code No: R07A1EC10 **R07** **Set No. 4**

I B.Tech Examinations,May 2011
COMPUTER PROGRAMMING AND NUMERICAL METHODS
Metallurgy And Material Technology

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

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1. What is the purpose of iterative statements? Explain about for loop with an example. [16]

2. (a) Find the positive root x^3- x = 1 correct to four decimal places by bisection method.

   (b) Find the parabola of the form y = ax^2 + bx + c passing through the points (0, 0), (1, 1) and (2, 2). [8+8]

3. (a) Using Lagrange's formula find f(6) given

X:	2	5	7	10	12
F(x)	18	180	448	1210	2028

   (b) If the interval of differencing is unity, prove that $\Delta \sin x = 2 \sin {}^1\!/_2 \cos \left(x + {}^1\!/_2\right)$. [8+8]

4. (a) How can a entire structure be passed to a function?

   (b) How can a entire structure be returned from a function? [8+8]

5. (a) Write an example to explain about call by value.

   (b) Write a factorial program using no return and no argument type. [10+6]

6. (a) By dividing the range in to five equal parts, evaluate $\int_0^{\pi} \sin x dx$ by Trapezoidal rule and Simpson's rule.

   (b) Using Milne's method find y (4.4) given $5xy' + y^2 - 2 = 0$ given y (4) =1, y(4.1) =1.0097, y(4.3) = 1.0143. [8+8]

7. (a) Write short notes on pointer to void.

   (b) Write short notes on Address Arithmetic. [8+8]

8. Define ADT. Explain with an example. [16]

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Code No: R07A1EC10 **R07** **Set No. 1**

I B.Tech Examinations,May 2011
COMPUTER PROGRAMMING AND NUMERICAL METHODS
Metallurgy And Material Technology

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

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1. (a) Find the positive root x^3- x = 1 correct to four decimal places by bisection method.
   (b) Find the parabola of the form y = ax^2 + bx + c passing through the points (0, 0), (1, 1) and (2, 2). [8+8]

2. (a) Write an example to explain about call by value.
   (b) Write a factorial program using no return and no argument type. [10+6]

3. (a) By dividing the range in to five equal parts, evaluate $\int_0^\pi \sin x dx$ by Trapezoidal rule and Simpson's rule.
   (b) Using Milne's method find y (4.4) given $5xy' + y^2 - 2 = 0$ given y (4) =1, y(4.1) =1.0097, y(4.3) = 1.0143. [8+8]

4. (a) Using Lagrange's formula find f(6) given

X:	2	5	7	10	12
F(x)	18	180	448	1210	2028

   (b) If the interval of differencing is unity, prove that $\Delta \sin x = 2 \sin {}^1\!/_2 \cos \left(x + {}^1\!/_2\right)$. [8+8]

5. (a) How can a entire structure be passed to a function?
   (b) How can a entire structure be returned from a function? [8+8]

6. Define ADT. Explain with an example. [16]

7. (a) Write short notes on pointer to void.
   (b) Write short notes on Address Arithmetic. [8+8]

8. What is the purpose of iterative statements? Explain about for loop with an example. [16]

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Code No: R07A1EC10 **R07** **Set No. 3**

I B.Tech Examinations,May 2011
COMPUTER PROGRAMMING AND NUMERICAL METHODS
Metallurgy And Material Technology

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

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1. Define ADT. Explain with an example. [16]

2. What is the purpose of iterative statements? Explain about for loop with an example. [16]

3. (a) How can a entire structure be passed to a function?
   (b) How can a entire structure be returned from a function? [8+8]

4. (a) Find the positive root x^3- x = 1 correct to four decimal places by bisection method.
   (b) Find the parabola of the form y = ax^2 + bx + c passing through the points (0, 0), (1, 1) and (2, 2). [8+8]

5. (a) Write an example to explain about call by value.
   (b) Write a factorial program using no return and no argument type. [10+6]

6. (a) Using Lagrange's formula find f(6) given

X:	2	5	7	10	12
F(x)	18	180	448	1210	2028

   (b) If the interval of differencing is unity, prove that $\Delta \sin x = 2 \sin {}^1\!/_2 \cos \left(x + {}^1\!/_2\right)$. [8+8]

7. (a) Write short notes on pointer to void.
   (b) Write short notes on Address Arithmetic. [8+8]

8. (a) By dividing the range in to five equal parts, evaluate $\int_0^{\pi} \sin x dx$ by Trapezoidal rule and Simpson's rule.
   (b) Using Milne's method find y (4.4) given $5xy' + y^2 - 2 = 0$ given y (4) =1, y(4.1) =1.0097, y(4.3) = 1.0143. [8+8]

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